Homework 3

MAT2040 : Summer 2023 : Prof. Lun Yu

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- Due date: 2023.6.30, 11.59 p.m.
- Late submission is NOT accepted.
- Please submit your answers as a PDF file with a name like "XXXXXXXXX HW3.pdf" (Your student ID + HW No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size $\leq$ 10MB).
- Please make sure that your submitted file is clear and readable. Submitted file that can not be opened or not readable will get 0 point.

Problem 1

Find the inverses of the matrices in (i) and (ii).

(i) $\begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & -2 & 1 \\ 4 & -7 & 3 \\ -2 & 6 & -4 \end{bmatrix}$

Problem 2

(i) Find invertible matrices A and B such that $A+B$ is not invertible.

(ii) Find singular matrices A and B such that $A+B$ is invertible.

Problem 3

Find the numbers a and b such that

$$\begin{bmatrix} 4 & -1 & -1 & -1 \\ -1 & 4 & -1 & -1 \\ -1 & -1 & 4 & -1 \\ -1 & -1 & -1 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} a & b & b & b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{bmatrix}.$$

Problem 4

Prove that A is invertible if $a \neq 0$ and $a \neq b$ (find the pivots or A^{-1}). Then find three numbers c so that C is not invertible:

$$A = \begin{bmatrix} a & b & b \\ a & a & b \\ a & a & a \end{bmatrix} \quad C = \begin{bmatrix} 2 & c & c \\ c & c & c \\ 8 & 7 & c \end{bmatrix}.$$

Problem 5

Let $A^N = 0$ for a certain integer $N > 0$, prove that $(I - A)^{-1} = I + A + A^2 + \cdots + A^{N-1}$.

Problem 6

(i) If $A = LDU$ and also $A = L_1 D_1 U_1$ with all factors invertible, prove that $L = L_1$, $D = D_1$ and $U = U_1$.

(ii) If $A = LU$ and also $A = L_1 U_1$ with all factors invertible, is $L = L_1$ and $U = U_1$ still true?

(iii) If $PA = LDU$ and also $P_1 A = L_1 D_1 U_1$ with all factors invertible, is $L = L_1$, $U = U_1, D = D_1$ and $P = P_1$ true?

Problem 7

Tridiagonal matrices have zero entries except on the main diagonal and the two adjacent diagonals. Factor these into $A = LU$ and $A = LDU$:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} a & a & 0 \\ a & a+b & b \\ 0 & b & b+c \end{bmatrix}.$$

Problem 8

Theorem of multiplicative property: If A and B are $n \times n$ matrices, then $det(AB)$=$det(A)det(B)$. Please verify this theorem for $A = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$

Problem 9

For each of the following, compute (i) det(A), and (ii)A^{-1}.

(i) $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ (ii) $A = \begin{bmatrix} 3 & 1 \\ 2 & 6 \end{bmatrix}$ (iii) $A = \begin{bmatrix} 3 & 3 & 1 \\ 2 & 1 & 1 \\ -2 & 2 & -1 \end{bmatrix}$ (iv) $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Problem 10

Compute the determinants of the following matrices.

(i)

$$A = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 7 & -1 & 0 & 0 \\ 2 & 6 & 3 & 0 \\ 7 & 2 & 1 & -3 \end{bmatrix}$$

(ii)

$$A = \begin{bmatrix} 7 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -6 & 5 & 7 & 2 \\ 2 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{bmatrix}$$

(iii)

$$A = \begin{bmatrix} 3 & 7 & 8 & 9 & -5 \\ 0 & 2 & -7 & 3 & 9 \\ 0 & 0 & 2 & 8 & 1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$$

(iv)

$$A = \begin{bmatrix} 8 & 2 & 1 & 0 & 2 \\ 2 & 0 & 1 & 3 & 4 \\ 1 & 5 & 2 & 2 & 0 \\ 0 & 4 & 9 & 2 & 0 \end{bmatrix}$$

Problem 11

Let A and B be 3×3 matrices, with det(A) = 3 and det(B) = 4. Use properties of determinants to compute:
(i) $det(AB)$
(ii) $det(5A)$
(iii) $det(B^T)$
(iv) $det(A^{-1})$
(v) $det(A^3)$

Problem 12

Let A be a $k \times k$ matrix and let B be an $(n-k) \times (n-k)$ matrix. Let $E = \begin{bmatrix} I_k & O \\ O & B \end{bmatrix}$, $F = \begin{bmatrix} A & O \\ O & I_{n-k} \end{bmatrix}$, $C = \begin{bmatrix} A & O \\ O & B \end{bmatrix}$ where I_k and I_{n-k} are the $k \times k$ and $(n-k) \times (n-k)$ identity matrices.
(i) Show that $det(E) = det(B)$.
(ii) Show that $det(F) = det(A)$.
(iii) Show that $det(C) = det(A)det(B)$.

Problem 13

Compute the determinants of the following matrices.
(i)

$$\begin{bmatrix} a & 1 & 0 & 0 \\ -1 & b & 1 & 0 \\ 0 & -1 & c & 1 \\ 0 & 0 & -1 & d \end{bmatrix}$$

(ii)

$$\begin{bmatrix} a & b & b & \dots & b \\ b & a & b & \dots & b \\ b & b & a & \dots & b \\ \dots & \dots & \dots & \dots & \dots \\ b & b & b & \dots & a \end{bmatrix}$$

(iii)

$$\begin{bmatrix} \alpha+\beta & \alpha\beta & 0 & 0 & \dots & 0 \\ 1 & \alpha+\beta & \alpha\beta & \cdots & 0 & 0 \\ 0 & 1 & \alpha+\beta & \cdots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & \alpha+\beta & \alpha\beta \\ 0 & 0 & 0 & \dots & 1 & \alpha+\beta \end{bmatrix}$$

Problem 14

Let A and B be 2×2matrices
$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$ $C = \begin{bmatrix} a_{11} & a_{12} \\ b_{21} & b_{22} \end{bmatrix}$ $D = \begin{bmatrix} b_{11} & b_{12} \\ a_{21} & a_{22} \end{bmatrix}$
Show that $det(A+B) = det(A) + det(B) + det(C) + det(D)$.

Problem 15

Let U be a square matrix such that $U^T U = I$. Show that $det(U) = \pm 1$.

Problem 16

Let a, b, c, $d \in \mathbb{R}$ and

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^3 & b^3 & c^3 & d^3 \end{bmatrix}$$

(i) Show that $det(A) = (d-a)(c-a)(b-a)(d-b)(c-b)(d-c)$.
(ii) Show a, b, c, d are distinct if and only if A is nonsingular.